

ELANTHIRAYAN MADHAVAN

Senior AI Engineer

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PROFESSIONAL SUMMARY

Senior AI Engineer with 11+ years of experience designing and delivering enterprise AI solutions, cloudnative applications, and scalable software platforms. Specialized in Generative AI, Agentic AI, Retrieval-Augmented Generation (RAG), Azure AI Foundry, Azure OpenAI, Vector Databases, and Large Language Model (LLM) applications. Proven expertise in architecting end-to-end AI systems, building production ready full-stack applications, and integrating intelligent solutions into enterprise workflows using Python, FastAPI, React, Next.js, and Microsoft Azure. Strong background in AI platform architecture, Software Architecture, and cloud-native engineering, complemented by experience in Digital Twins, OpenUSD, and XR/Spatial Computing. Active open-source contributor, technical writer, and creator of AI Concepts Playground, an interactive platform that simplifies AI concepts through hands-on learning.

PROFESSIONAL EXPERIENCE

Shell India Markets Pvt Ltd | 2018 - Present

L&T Technology Services (2018–2023), Randstad India Pvt Ltd (2023–2025), Shell India Markets Pvt Ltd – Direct Employee (2025–Present)

Innovation Analyst (AI, HPC & Spatial Computing)

Design and develop enterprise AI solutions, cloud-native applications, Digital Twin platforms, and intelligent software systems for global business initiatives. Collaborate with cross-functional teams to architect scalable AI applications, modern software platforms, and emerging technology solutions.

Key Contributions

Artificial Intelligence & Generative AI

- Architected and delivered enterprise Retrieval-Augmented Generation (RAG) platforms for patent and technical document search using Python, FastAPI, Azure AI Foundry, Azure OpenAI, LlamaIndex, FAISS, and Next.js, enabling natural language search across enterprise knowledge repositories.
- Engineered scalable document intelligence pipelines for parsing, chunking, metadata extraction, embedding generation, vector indexing, and hybrid retrieval using BM25, semantic search, and re-ranking.
- Accelerated large-scale document processing by leveraging High-Performance Computing (HPC) to parallelize extraction, preprocessing, embedding generation, and indexing workloads.
- Orchestrated LangGraph-based multi-agent workflows for planning, retrieval, reasoning, validation, and tool calling to answer complex engineering queries.
- Integrated Azure OpenAI and embedding models into production applications, delivering semantic search, contextual retrieval, summarization, and conversational AI capabilities.
- Collaborated with cross-functional stakeholders to design and deliver scalable AI solutions for enterprise knowledge management and intelligent document processing.

Software Architecture & Cloud Engineering

- Designed scalable backend architectures and REST APIs using Python, FastAPI, and Microsoft Azure, delivering secure, maintainable, and cloud-native enterprise applications.
- Developed and maintained cloud-native solutions using Azure App Services, Azure Functions, Storage Accounts, Key Vault, CI/CD pipelines, and GitHub Actions, supporting production deployments and operational reliability.
- Collaborated with business stakeholders, global engineering teams, and technology partners to design solution architectures, evaluate technologies, and successfully deliver multiple enterprise initiatives.

Digital Twins, OpenUSD & Spatial Computing

- Authored a technical research paper proposing an enterprise adoption strategy for OpenUSD, providing architectural guidance, interoperability recommendations, and implementation considerations for future Digital Twin and 3D engineering initiatives.

- Architected a Digital Twin visualization platform that transformed operational data into immersive 3D environments, improving operational situational awareness by 72%.
- Developed a spatial computing application for Apple Vision Pro using Unity and SwiftUI, enabling real-time visualization of Digital Twin operational alerts and monitoring information. Built proof-of-concepts combining Machine Vision, Generative AI, XR, and voice interaction, enabling AI-assisted creation and visualization of 3D assets within immersive environments.

Technical Leadership & Innovation

- Authored internal technical papers covering OpenUSD, CloudXR, Digital Twins, XR technologies, and emerging engineering practices, supporting technology evaluation and enterprise innovation initiatives.
- Researched, evaluated, and demonstrated emerging AI, XR, and Digital Twin technologies through proof-of-concepts, contributing to future technology roadmaps and engineering best practices.

ASM Technologies Ltd | 2017 - 2018

Full-Stack Software Engineer

Developed enterprise web applications, backend services, and immersive XR training solutions for automotive and semiconductor industries, integrating real-time visualization with cloud-based business systems.

Key Contributions

- Developed immersive VR training applications using Unity for automotive and semiconductor assembly/disassembly processes, reducing training time by 65% compared to conventional methods.
- Built interactive ADAS simulation environments, enabling engineers to visualize, configure, and evaluate vehicle systems from multiple perspectives, improving technical understanding by 40%. Designed and implemented REST APIs connecting XR applications with enterprise backend systems, enabling secure data exchange and real-time synchronization.
- Developed web-based administration portals for content management, user analytics, and training reporting using modern web technologies.
- Collaborated with cross-functional engineering teams to deliver enterprise XR solutions for manufacturing and industrial training.

Axinovate Technologies LLP | 2015 - 2017

Full-Stack Software Engineer

Designed and developed full-stack web applications, real-time collaboration platforms, and cloudhosted enterprise solutions across education and event management domains.

Key Contributions

- Developed full-stack features for Skilladda and Xucore e-learning platforms, contributing across frontend, backend, database, and deployment.
- Designed and developed a real-time multiplayer event management platform using Node.js, WebRTC, and MySQL, enabling live audience participation at NVIDIA Gamer Connect events across Asia with 95% live quiz participation.
- Built hybrid mobile applications for Android and iOS using Apache Cordova, extending platform accessibility across mobile devices.
- Managed cloud deployments using Microsoft Azure App Services and Azure Blob Storage, supporting scalable application hosting and content delivery.
- Collaborated with stakeholders throughout the software development lifecycle, delivering production-ready web applications using modern full-stack development practices. Technologies: Node.js, PHP, JavaScript, MySQL, Apache Cordova, Microsoft Azure

OPEN SOURCE CONTRIBUTIONS

AI Concepts Playground (*Creator & Maintainer*) - Open-source interactive AI learning platform with 50+ hands-on lessons covering LLMs, RAG, MCP, AI Agents, and LLM Architecture. Designed interactive visualizations and browser-based experiments to simplify complex AI concepts for developers. <https://imelanthirayan.github.io/aip playground/>

MilvusLite Kit - Built a lightweight Python toolkit simplifying vector database integration for Retrieval-Augmented Generation (RAG) applications.

Chat UI - Developed a configurable AI chat platform supporting multiple LLM providers and extensible backend integrations.

FolderWatcher - Event-driven Python framework that monitors filesystem changes and automates document ingestion workflows for AI and RAG pipelines.

EXTERNAL PUBLICATIONS

- Published a research paper on Reinforcement Learning for Robotic Simulation in collaboration with IIM Bangalore.
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EDUCATION

M.Tech in Data Science and Machine Learning | 2022

PES UNIVERSITY

Master of Computer Applications | 2018

ANNAMALAI UNIVERSITY

TECHNICAL SKILLS

- Generative AI & Agentic AI: LLMs, Retrieval-Augmented Generation (RAG), LangGraph, AI Agents, Microsoft Foundry (Azure AI Foundry), Azure OpenAI, Knowledge Graphs, Neo4j
- Cloud & DevOps: Microsoft Azure, Azure Storage Accounts, Docker, Azure App Services, GitHub Actions, CI/CD, High-Performance Computing (HPC)
- Programming: Python, C#, JavaScript, SQL
- Frameworks: FastAPI, React, Next.js, Node.js
- Spatial Computing: OpenUSD, Digital Twins, Unity, Apple Vision Pro, XR